Ajay Devidas Wanekar

M.Tech Energy Science and Technology - TA

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SUMMARY

M.Tech student at IIT Hyderabad with 2+ years of research experience in Artificial Intelligence and Machine Learning, specializing in Computer Vision, OCR-based text and character recognition, and Vision-Language Models. Experienced in image preprocessing, dataset annotation, model benchmarking, and matching extracted text against reference data.

Degree	University/Institute	Year	CGPA/ Marks(%)
M.Tech Energy Science and Technology - TA	IIT Hyderabad	2027	6.8
B.E. Electrical Engineering	Savitribai Phule Pune University	2022	7.28
XII (Maharashtra State Board)	Vasantrao Naik Jr. College of Science and Arts, Cidco, Ch. Sambhajinagar, Maharashtra	2017	67.38%
X (Maharashtra State Board)	Maharashtra Public School, DTP, Ch. Sambhajinagar	2014	83.00%

WORK EXPERIENCE

Project Associate | LFOVIA Lab, IIT Hyderabad Sep 2024 – Jul 2025

Evaluated and fine-tuned Vision-Language Models (VLMs), including Florence-2 Base/Large, Llama 3.2 Vision, and Qwen2-VL, for multimodal text-image understanding and OCR-based text and character recognition, and benchmarked task-specific models (RetinaFace, DeepFace, Siamese networks) for detection, face analysis, and visual tracking.

Junior Research Assistant | TIHAN, IIT Hyderabad: Nov 2023 – Jul 2024

Worked on multimodal maritime perception by developing and curating near-infrared and sonar datasets, implementing sensor-specific preprocessing, annotation, data-quality optimization, and model-training pipelines for robust object detection in complex maritime environments, resulting in an IEEE conference publication.

Research Intern | WINET Lab, IIT Hyderabad: Jul 2023 – Nov 2023

Developed foundational sonar-based underwater object-detection datasets by implementing annotation, Python-based preprocessing, data augmentation, and markup pipelines to improve data quality and model robustness, with the work published at the IEEE International Conference on Sustainable Technology and Engineering (i-COSTE 2023).

PROJECTS

Asbestos-Cement Roof Detection from Satellite Data (open source, GitHub):

- Classified asbestos-cement roofs from 8-band satellite spectra, trained in Australia and tested in an unseen region: 95.2% accuracy (F1 0.955) vs 90.2% for the dataset authors' best model, with no labels from the new region.
- Spectral-shape features (band ratios and normalised differences) made the ranking robust to scene brightness (AUC 0.984), and a threshold fitted to the new region's unlabelled scores fixed the shift that cut standard models to 80 to 91%.

Vision-Language Model Evaluation for Computer Vision and OCR Tasks (LFOVIA Lab, IIT Hyderabad):

- Evaluated Vision-Language Models (VLMs) for multimodal (text-image) understanding across computer vision tasks, including OCR-based text and character recognition, object detection, segmentation, age and gender classification, crowd analysis, and action/activity recognition, as alternatives to conventional task-specific models.
- Curated and annotated RGB and depth-mapping datasets with preprocessing, labeling, and quality-validation pipelines.
- Benchmarked and fine-tuned models using Python-based pipelines, evaluating detection and recognition performance (precision/recall, IoU).

Multi-Channel Conversational AI Platform (IIT Hyderabad Build Program):

- Built a multi-tenant conversational AI platform (voice, chat, email; real-time STT/TTS, multilingual) with a hybrid semantic + keyword retrieval pipeline that matches noisy text against a reference knowledge base, improving answer relevance by 18%.

THESIS PROJECT

Machine Learning for Non-Oxidative Methane Coupling in a DBD Reactor

- Developed and benchmarked Linear Regression, Random Forest, XGBoost, and Gaussian Process Regression models to predict reactor performance from operating conditions, validated with Leave-One-Out Cross-Validation (LOOCV), reaching cross-validated R^2 up to 0.996 across six reactor-performance outputs, and identified Bayesian Optimization for experiment-guided process optimization.
- Recently adapted the RRSI method to this work to rigorously verify that model improvements are real.

TECHNICAL SKILLS

Programming & Tools: Python, C++, SQL, PyTorch, TensorFlow, OpenCV, CUDA, NumPy, Pandas, Scikit-learn

Computer Vision & OCR: OCR (Text and Character Recognition), Image Preprocessing, Object Detection, Data Augmentation, Dataset Annotation, CNN, YOLO, DeepFace, RetinaFace

Multimodal AI & LLMs: Vision-Language Models (Florence-2, Qwen2-VL, Llama 3.2 Vision), Transformer Architectures, Large Language Models (LLMs), LLM Fine-Tuning, RAG, Hybrid Semantic + Keyword Retrieval

Model Evaluation: Precision, Recall, IoU, Model Benchmarking

CONFERENCE PUBLICATIONS

[1] Wanekar A. D., Mannam N. P. B., Rajalakshmi P. "Novel Approach to Underwater Object Detection Using Sonar Sensors for Autonomous Underwater Vehicles (AUVs)." IEEE i-COSTE 2023, pp. 1–5.

[2] Wanekar A. D., Mannam N. P. B., Rajalakshmi P. "Object Detection and Classification for Autonomous Surface Vehicles (ASVs) Through Near-Infrared Imaging." IEEE i-COSTE 2023, pp. 1–6.

HACKATHONS, ACHIEVEMENTS & LANGUAGES

- Top 10 All India - Winner, Meta x Pragati AI Hackathon.
- AI and X Hackathon, IIT Hyderabad participant

Languages: English, Hindi, Marathi, Japanese (Basic)